

CSEC Mathematics
June 2014 – Paper 2
Solutions

SECTION I

Answer ALL questions in this section.

All working must be clearly shown.

1. (a) Using a calculator, or otherwise, determine the value of:

(i) $5.25 \div 0.015$ [1]

Using a calculator,

$$5.25 \div 0.015 = 350$$

(ii) $\sqrt{6.5025}$ [1]

Using a calculator,

$$\sqrt{6.5025} = 2.55$$

(iii) 3.142×2.236^2 (correct to 3 significant figures) [2]

Using a calculator,

$$3.142 \times 2.236^2 = 15.7 \text{ (to 3 significant figures)}$$

(b) Concrete tiles are made using buckets of cement, sand and gravel mixed in the ratio 1 : 4 : 6.

(i) How many buckets of gravel are needed for 4 buckets of cement? [2]

Cement : Sand : Gravel

1 : 4 : 6

($\times 4$)

4 : 16 : 24

$\therefore$ 4 buckets of cement would require 24 buckets of gravel.

(ii) If 20 buckets of sand are used, how many buckets of EACH of the following will be needed?

(a) Cement

Cement : Sand : Gravel

1 : 4 : 6

($\times 5$)

5 : 20 : 30

$\therefore$ 5 buckets of cement will be needed.

(b) Gravel

1 bucket of cement is to be mixed with 6 buckets of gravel

5 buckets of cement will be mixed with $5 \times 6 = 30$ buckets of gravel

$\therefore$ 30 buckets of gravel will be needed.

(c) The cash price of a laptop is \$1 299. It can be bought on hire purchase by making a deposit of \$350 and 10 monthly payments of \$120 each.

(i) What is the TOTAL hire purchase price of the laptop? [2]

Hire Purchase Price = Deposit + Instalments

$$= \$350 + (10 \times \$120)$$

$$= \$350 + \$1200$$

$$= \$1550$$

$\therefore$ The total hire purchase price of the laptop is \$1550.

(ii) How much is saved by buying the laptop for cash? [1]

Amount saved = Hire Purchase Price – Cash Price

$$= \$1550 - \$1299$$

$$= \$251$$

$\therefore$ \$251 is saved by buying the laptop for cash.

Total: 12 marks

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2. (a) Write as a single fraction in its LOWEST terms

$$\frac{x-2}{3} + \frac{x+1}{4}$$

[3]

$$\begin{aligned} & \frac{x-2}{3} + \frac{x+1}{4} \\ &= \frac{4(x-2)+3(x+1)}{12} \\ &= \frac{4x-8+3x+3}{12} \\ &= \frac{7x-5}{12} \end{aligned}$$

- (b) Write an equation in x to represent EACH statement below. Do NOT solve the equation.

- (i) When 4 is added to a certain number the result is the same as halving the number and adding 10. [1]

Let the unknown number be x .

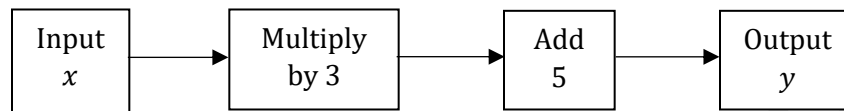
An equation to represent the statement is: $4 + x = \frac{1}{2}x + 10$

- (ii) Squaring a number and subtracting 6 gives the same result as doubling the number and adding 9. [1]

Let the unknown number be x .

An equation to represent the statement is: $x^2 - 6 = 2x + 9$

(c) John drew the diagram below to show what he was thinking.



- (i) Use information from the diagram to write a formula for y in terms of x . [1]

A formula for y in terms of x is $y = 3x + 5$.

- (ii) If the number 4 is the input, what number would be the output? [1]

When $x = 4$,

$$y = 3(4) + 5$$

$$y = 12 + 5$$

$$y = 17$$

- (iii) If the number 8 was the output, what number was the input? [1]

When $y = 8$,

$$3x + 5 = 8$$

$$3x = 8 - 5$$

$$3x = 3$$

$$x = \frac{3}{3}$$

$$x = 1$$

- (iv) Reverse the formula written at (c)(i) above to write x in terms of y . [1]

$$y = 3x + 5$$

$$3x = y - 5$$

$$x = \frac{y-5}{3}$$

- (d) Solve the following simultaneous equations:

$$2x + 3y = 9$$

$$3x - y = 8 \quad [3]$$

$$2x + 3y = 9 \quad \dots \text{Equation 1}$$

$$3x - y = 8 \quad \dots \text{Equation 2}$$

Multiplying Equation 2 by 3 gives,

$$9x - 3y = 24 \quad \dots \text{Equation 3}$$

$$2x + 3y = 9 \quad \dots \text{Equation 1}$$

Adding Equation 1 and Equation 3 gives,

$$11x = 33$$

$$x = \frac{33}{11}$$

$$x = 3$$

Substituting $x = 3$ into Equation 1 gives,

$$2(3) + 3y = 9$$

$$6 + 3y = 9$$

$$3y = 9 - 6$$

$$3y = 3$$

$$y = \frac{3}{3}$$

$$y = 1$$

$$\therefore x = 3 \text{ and } y = 1$$

Total: 12 marks

3. (a) The universal set, U , is defined as the set of integers between 11 and 26.

A and B are subsets of U such that:

$$A = \{\text{even numbers}\}$$

$$B = \{\text{multiples of 3}\}$$

- (i) How many members are in the universal set, U ? [1]

$$\begin{aligned} U &= \{\text{set of integers between 11 and 26}\} \\ &= \{11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26\} \end{aligned}$$

Now,

$$n(U) = 16$$

$\therefore$ 16 members are in the universal set, U .

- (ii) List the members of the subset A . [1]

$$A = \{\text{even numbers}\}$$

$$A = \{12, 14, 16, 18, 20, 22, 24, 26\}$$

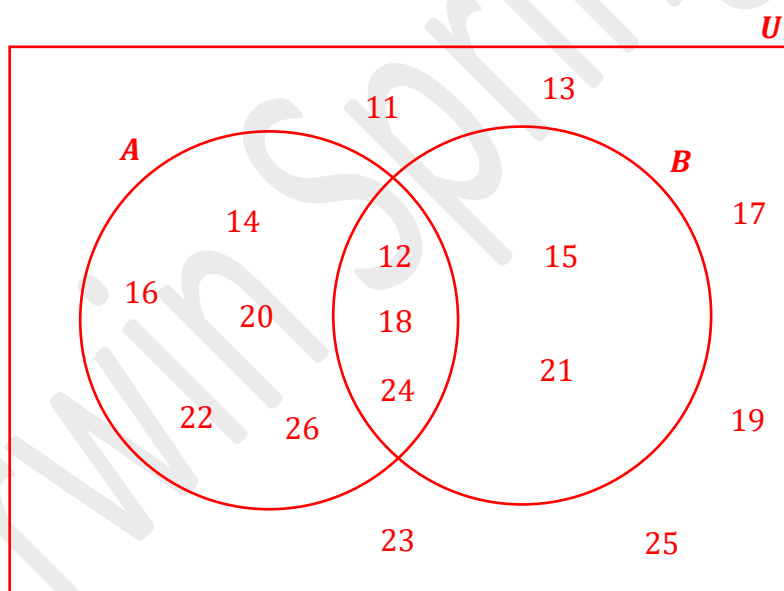
- (iii) List the members of the subset B .

$$B = \{\text{multiples of } 3\}$$

$$= \{12, 15, 18, 21, 24\}$$

- (iv) Draw a Venn diagram to represent the relationships among A , B and U . [3]

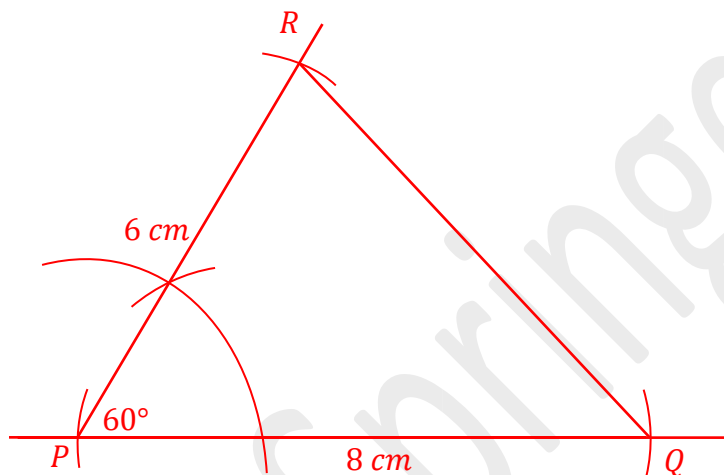
The Venn diagram is shown below:



(b) (i) Using a ruler, a pencil and a pair of compasses, construct

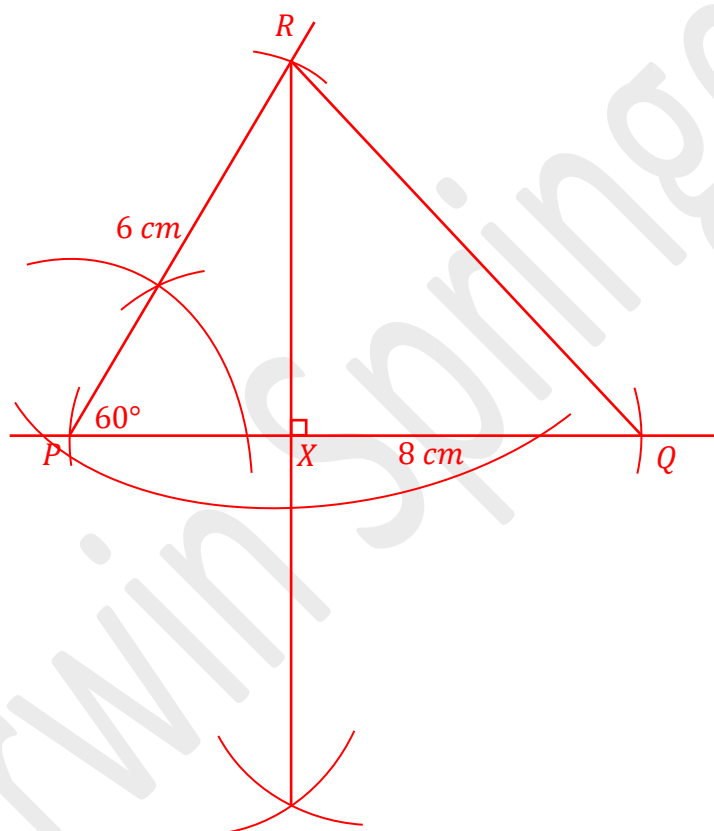
(a) a triangle PQR in which $PQ = 8$ cm, $PR = 6$ cm and angle $P = 60^\circ$ [3]

The construction diagram is shown below:



- (b) the line segment RX which is perpendicular to PQ and meets PQ at X . [2]

The construction diagram is shown below:



[Note: Credit will be given for clearly drawn construction lines.]

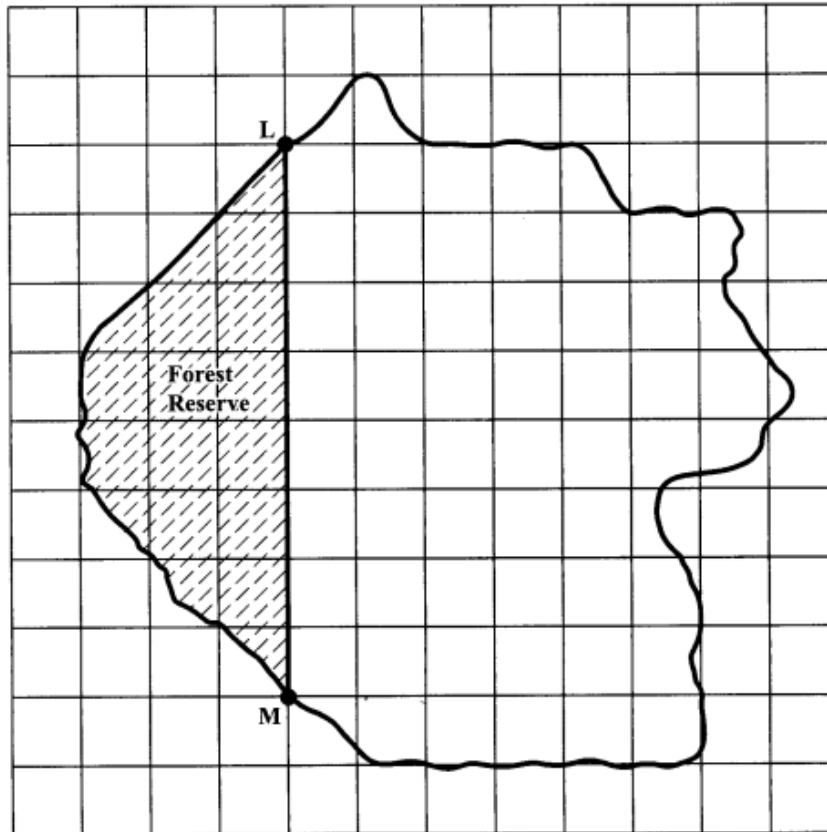
(ii) Measure and state the size of angle QRX .

With the use of a protractor, the size of $\angle QRX = 44^\circ$.

Total: 12 marks

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4. The diagram below shows a map of an island drawn on a grid of 1-*cm* squares. The map is drawn to a scale of 1: 50 000.



(a) Copy and complete EACH of the following sentences:

- (i) 1 *cm* on the map represents 50 000 *cm* on the island. [1]

The scale given is 1: 50 000

$\therefore$ 1 *cm* on the map represents 50 000 *cm* on the island

- (ii) An area of 1 cm^2 on the map represents an area of $\underline{2.5 \times 10^9} \text{ cm}^2$ on the island. [1]

Since 1 cm on the map represents $50\,000 \text{ cm}$ on the island,

Then 1 cm^2 on the map represents $(50\,000)^2 \text{ cm}^2$ on the island.

$$= 2\,500\,000\,000 \text{ cm}^2 \text{ on the island.}$$

$$= 2.5 \times 10^9 \text{ cm}^2 \text{ on the island.}$$

- (iii) Given that $1 \text{ km} = 100\,000 \text{ cm}$, a distance of 1 cm on the map represents a distance of $\underline{\frac{1}{2}}$ km on the island. [1]

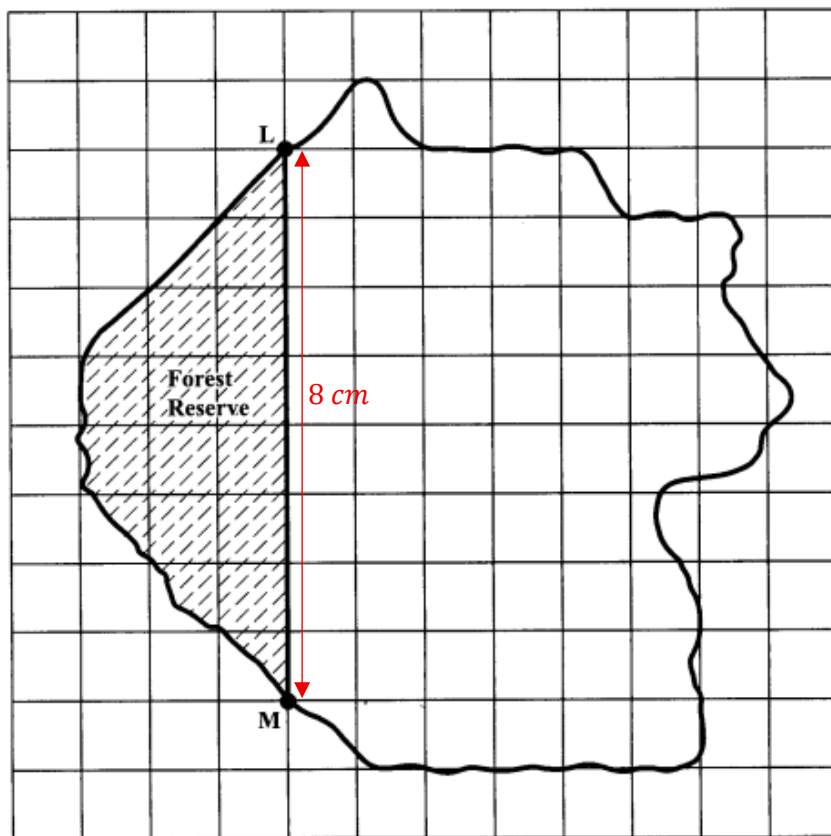
$$1 \text{ cm on the map} = 50\,000 \text{ cm}$$

$$= \frac{50\,000}{100\,000} \text{ km}$$

$$= \frac{1}{2} \text{ km}$$

- (b) (i) L and M are two tracking stations. State, in centimetres, the distance LM on the map.

[1]



The distance of L to M is 8 cm on the map.

$\therefore LM = 8 \text{ cm}$ on the map

- (ii) Calculate the ACTUAL distance, in kilometres, from L to M on the island. [2]

$$1 \text{ cm} = \frac{1}{2} \text{ km}$$

$$\begin{aligned} \text{The actual distance } LM &= \left(8 \times \frac{1}{2}\right) \text{ km} \\ &= 4 \text{ km} \end{aligned}$$

- (c) (i) The area shaded on the map is a forest reserve. By counting squares estimate, in cm^2 , the area of the forest reserve as shown on the map. [2]

The total number of squares is approximately 15.

Hence,

$$\begin{aligned} \text{Estimated area of the forest reserve on the map} &= 15 \times 1 \\ &= 15 \text{ cm}^2 \end{aligned}$$

- (ii) Calculate, in km^2 , the ACTUAL area of the forest reserve. [2]

$$1 \text{ cm} = \frac{1}{2} \text{ km}$$

$$1 \text{ cm}^2 = \left(\frac{1}{2}\right)^2 \text{ km}^2$$

$$= \frac{1}{2} \times \frac{1}{2} \text{ km}^2$$

$$= \frac{1}{4} \text{ km}^2$$

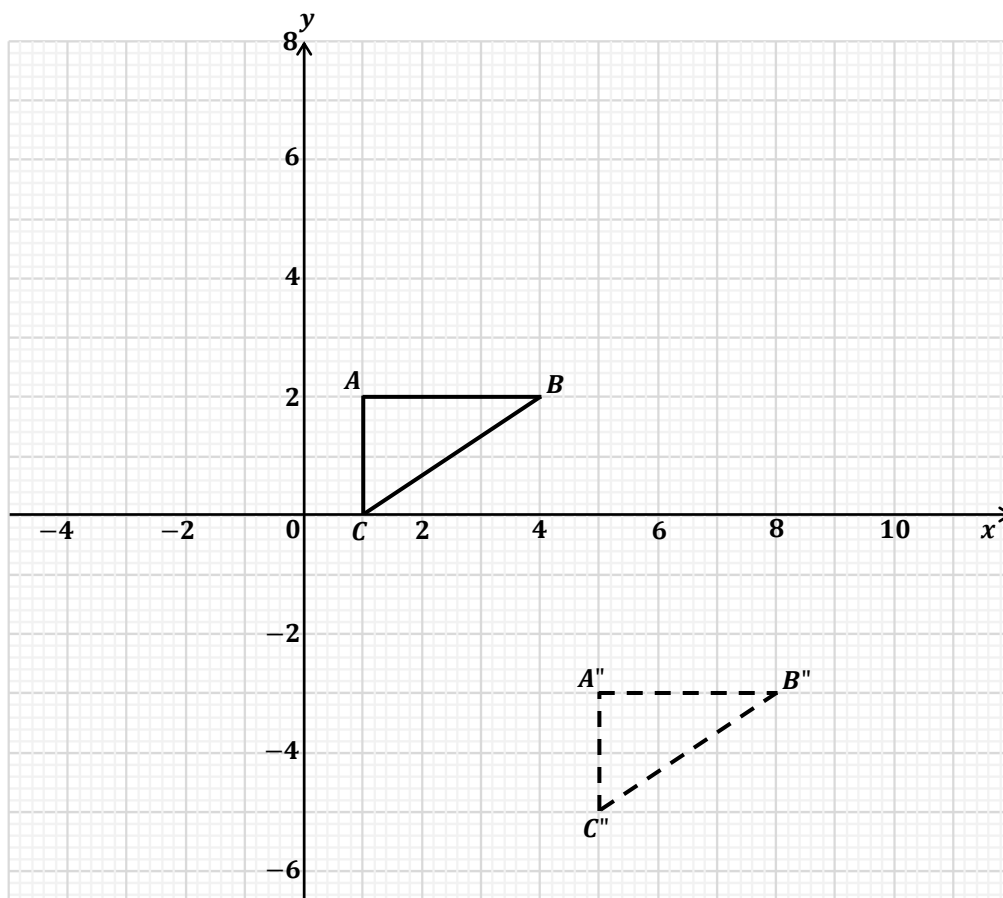
Actual area of the forest reserve on the map = $15 \times \left(\frac{1}{4}\right) \text{ km}^2$

$$= \frac{15}{4} \text{ km}^2$$

$$= 3 \frac{3}{4} \text{ km}^2$$

Total: 10 marks

5. An answer sheet is provided for this question.



(a) Triangle ABC has coordinates $A(1, 2)$, $B(4, 2)$ and $C(1, 0)$.

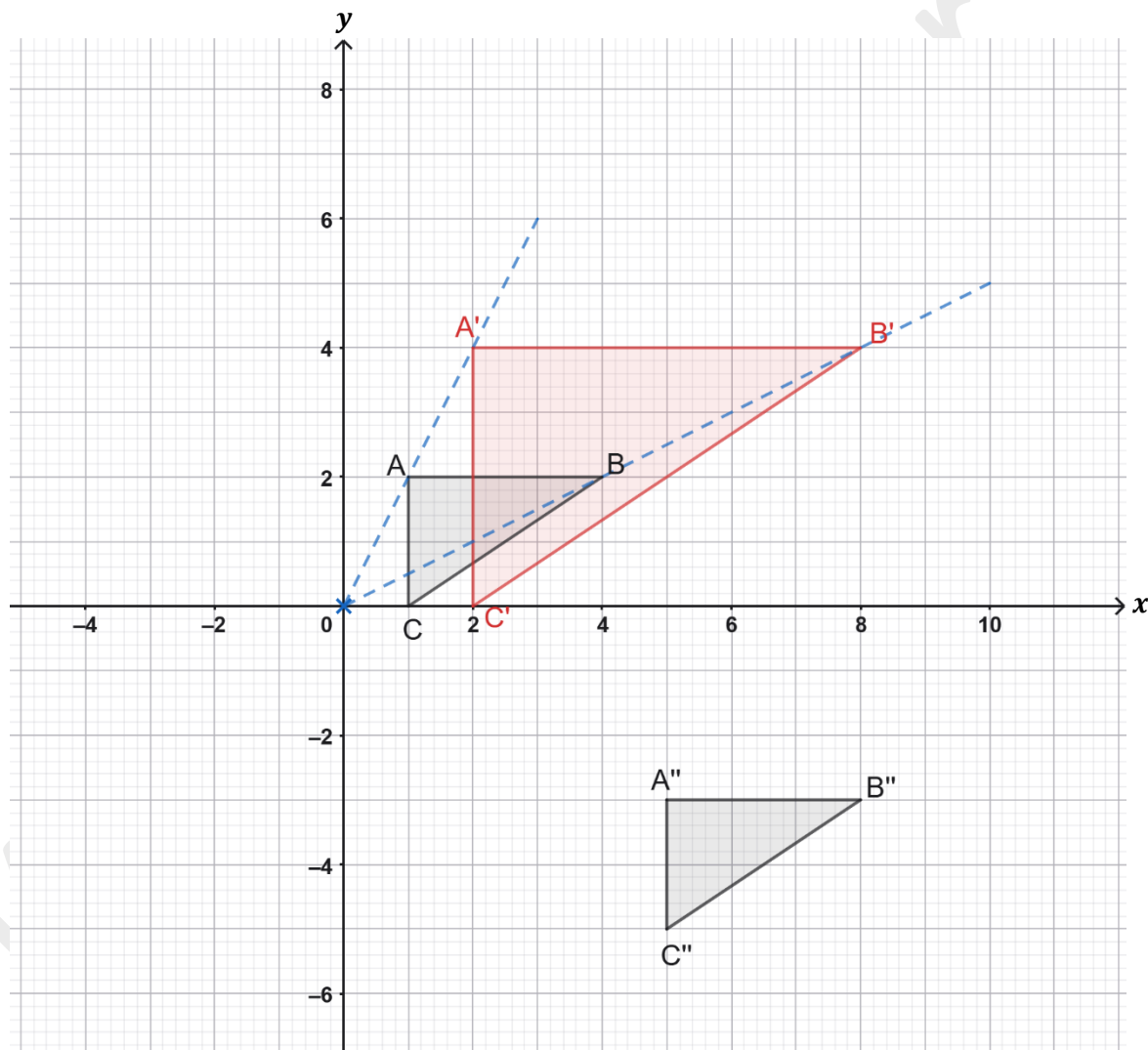
- (i) **On the answer sheet provided**, draw triangle $A'B'C'$, the image of triangle ABC , under an enlargement, centre O and scale factor 2. [3]

Triangle $A'B'C'$ is drawn below.

$$A' = (2, 4)$$

$$B' = (8, 4)$$

$$C' = (2, 0)$$



- (ii) Triangle $A''B''C''$ is the image of triangle ABC , under a transformation, M .

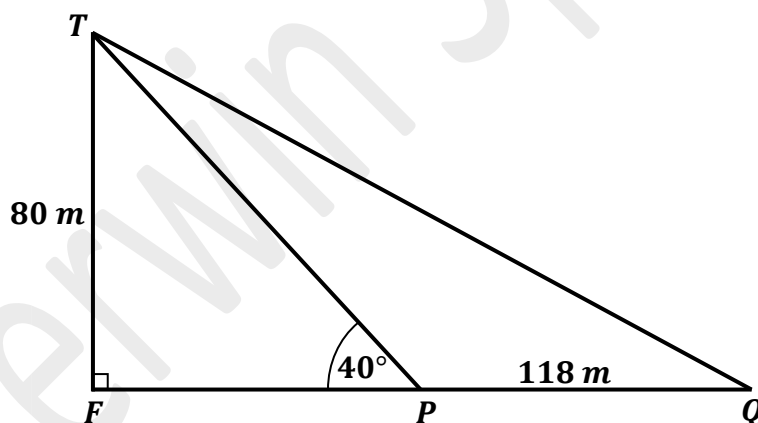
Describe **completely** the transformation, M .

[3]

The triangle ABC is mapped onto triangle $A''B''C''$ by a shift of 4 units horizontally to the right and 5 units vertically downwards and is a translation.

$\therefore M$ is a translation, represented by the matrix, $M = \begin{pmatrix} 4 \\ -5 \end{pmatrix}$.

- (b) The diagram below, not drawn to scale, shows the position of two ships, P and Q , at anchor. FT is the vertical face of a cliff jutting out of the water. P and Q are 118 m apart. $FT = 80$ m and $\angle FPT = 40^\circ$.



Determine

- (i) the angle of elevation of T from P

[1]

The angle of elevation, shown by the angle $F\hat{P}T = 40^\circ$.

(ii) the length of FP

$$\tan 40^\circ = \frac{80}{FP}$$

$$\therefore FP = \frac{80}{\tan 40^\circ}$$

$$= 95.34 \text{ m}$$

(iii) the angle of elevation of T from Q

[3]

$$\tan T\hat{Q}F = \frac{80}{95.34+118}$$

$$\tan T\hat{Q}F = 0.375$$

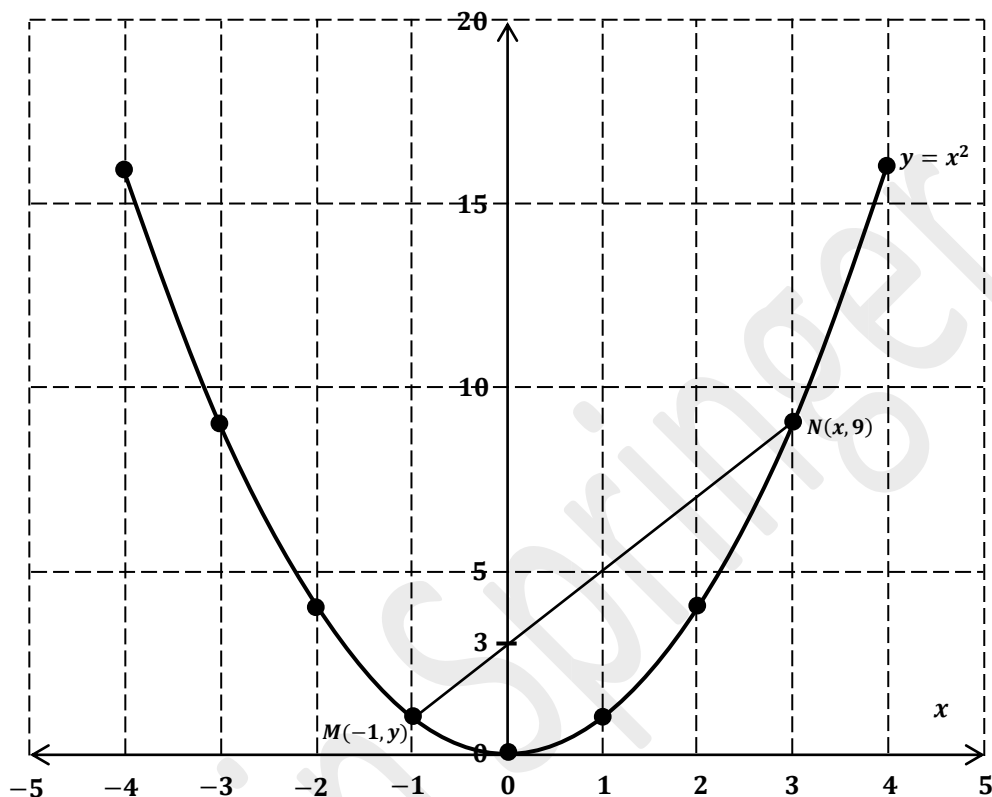
$$T\hat{Q}F = \tan^{-1}(0.375)$$

$$= 20.6^\circ \quad (\text{to 1 decimal place})$$

Total: 12 marks

6. An answer sheet is provided for this question.

The graph of the quadratic function $y = x^2$ for $-4 \leq x \leq 4$ is shown below.



(a) The coordinates of the points M and N are $(-1, y)$ and $(x, 9)$ respectively.

Determine the value of

(i) x

[1]

$$N(x, 9)$$

$$y = x^2$$

$$\therefore 9 = x^2$$

$$x = \sqrt{9}$$

$$x = \pm 3$$

Since $x > 0$, $x = 3$.

(ii) y [1]

$$M(-1, y)$$

$$y = x^2$$

$$\therefore y = (-1)^2$$

$$y = 1$$

(b) Determine

(i) the gradient of the line MN [1]

$$M = (-1, 1) \text{ and } N = (3, 9)$$

$$\text{Gradient of } MN = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{9 - 1}{3 - (-1)}$$

$$= \frac{8}{4}$$

$$= 2$$

- (ii) the equation of the line MN

$$\text{Gradient of } MN = 2$$

Using point $N = (3, 9)$,

$$\frac{y-9}{x-3} = 2$$

$$y - 9 = 2(x - 3)$$

$$y - 9 = 2x - 6$$

$$y = 2x - 6 + 9$$

$$y = 2x + 3$$

- (iii) the equation of the line parallel to MN, and passing through the origin. [2]

Since parallel line have the same gradient, the required line has the same gradient as MN , which is 2. The origin is $(0, 0)$.

$\therefore$ Equation of the required line is $\frac{y-0}{x-0} = 2$.

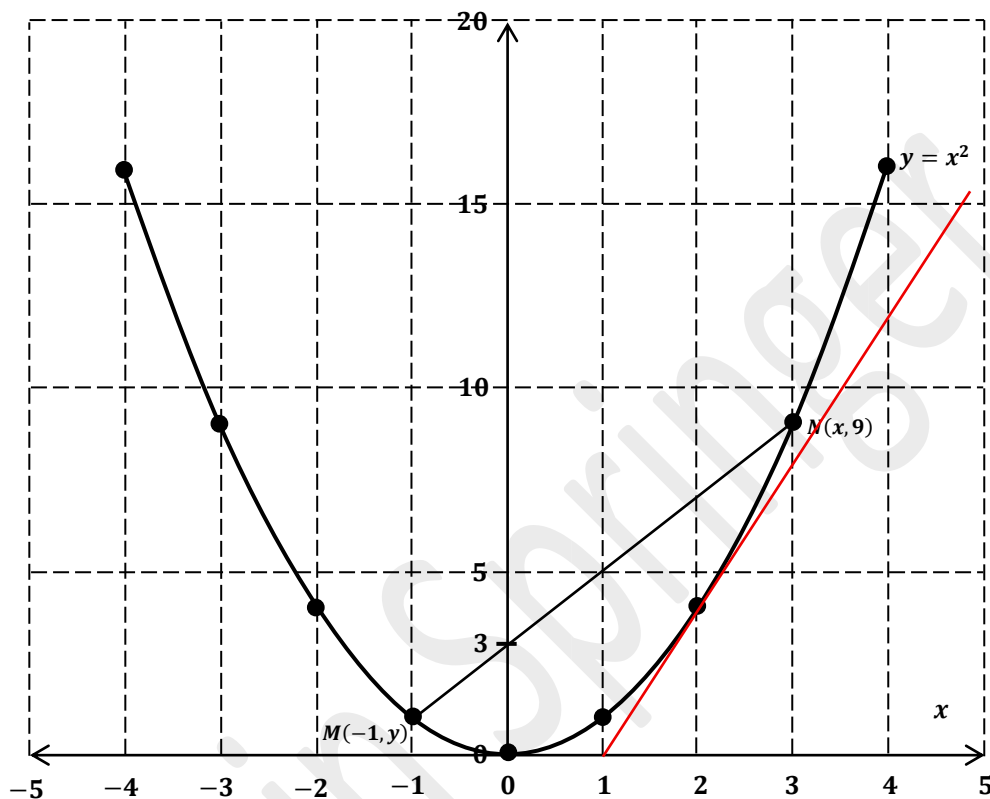
$$\frac{y}{x} = 2$$

$$y = 2x$$

(c) On the answer sheet provided, carefully, draw the tangent line to the graph

$$y = x^2 \text{ at the point } (2, 4).$$

[2]



(d) Estimate the gradient of the tangent to the curve at (2, 4).

[2]

Using points (2, 4) and (1, 0),

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{4 - 0}{2 - 1}$$

$$m = \frac{4}{1}$$

$$m = 4$$

Total: 11 marks

7. A class of 30 students counted the number of books in their bags on a certain day.

The number of books in EACH bag is shown below.

5 4 6 3 2 1 7 4 5 3
 6 5 4 3 7 6 2 5 4 5
 5 7 5 4 3 2 1 6 3 4

- (a) Copy and complete the frequency table for the data shown above. [4]

Number of Books (x)	Tally	Frequency (f)	$f \times x$
1		2	2
2		3	6
3		5	15
4	I	6	24
5	II	7	35
6		4	24
7		3	21

- (b) State the modal number of books in the bags of the sample of students. [1]

The highest frequency is 7 which corresponds to 5 books. Therefore, the modal number of books is 5 books.

(c) Using the table in (a) above, or otherwise, calculate

- (i) the TOTAL number of books [2]

$$\begin{aligned}\sum fx &= 2 + 6 + 15 + 24 + 35 + 24 + 21 \\ &= 127\end{aligned}$$

$\therefore$ The total number of books is 127 books.

- (ii) the mean number of books per bag. [2]

$$\begin{aligned}\bar{x} &= \frac{\sum fx}{\sum f} \\ &= \frac{127}{30} \\ &= 4.23 \text{ books (to 3 significant figures)}\end{aligned}$$

$\therefore$ The mean number of books per bag is 4.23 books.

- (d) Determine the probability that a student chosen at random has LESS THAN 4 books in his/her bag. [2]

$$\begin{aligned}\text{Probability} &= \frac{\text{Number of students who have less than 4 books}}{\text{Total number of students}} \\ &= \frac{2+3+5}{30} \\ &= \frac{10}{30} \\ &= \frac{1}{3}\end{aligned}$$

Total: 11 marks

8. An answer sheet is provided for this question.

A number sequence may be formed by counting the number of dots used to draw each set of geometric figures. The first three figures are shown below.

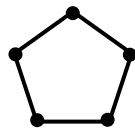


Figure 1

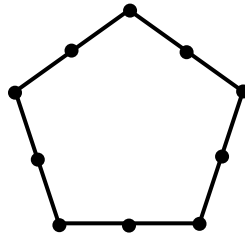


Figure 2

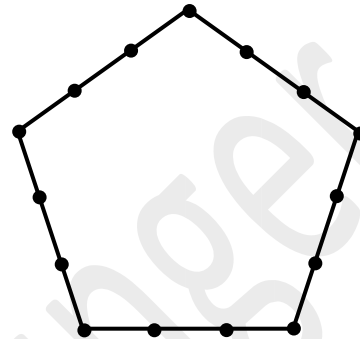


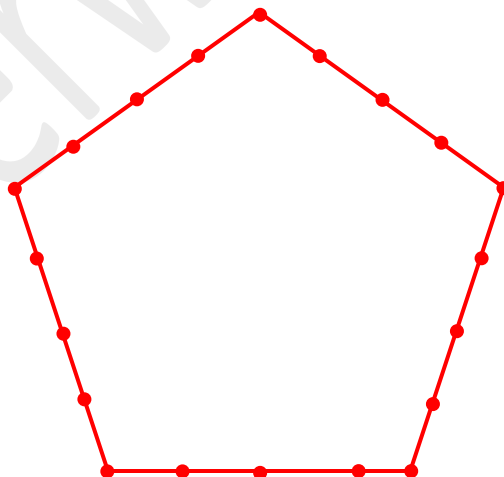
Figure 3

On the answer sheet provided,

(a) Draw Figure 4, the next figure in the sequence above.

[2]

Figure 4 is shown below:



(b) Complete the table below by inserting the missing information at the rows

numbered (i) and (ii).

[4]

Figure (f)	Total Number of Dots	
	Formula	Number (n)
1	$5 \times 2 - 5$	5
2	$5 \times 3 - 5$	10
3	$5 \times 4 - 5$	15
4	Not required	Not required
(i) 5	$5 \times (5 + 1) - 5$	25
(ii) 6	$5 \times (6 + 1) - 5$	30

(c) Write an expression in f for the number (n) of dots used in drawing the f th figure.

[2]

The number of dots, n , in the f th figure is given by: $5f = n$

(d) Which figure in the sequence contains 145 dots?

[2]

$$145 = 5f$$

$$f = \frac{145}{5}$$

$$f = 29$$

Hence, there are 145 dots in the 29th figure.

Total: 10 marks

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SECTION II

Answer TWO questions in this section.

ALGEBRA AND RELATIONS, FUNCTIONS AND GRAPHS

9. (a) Two functions are defined as follows:

$$g(x) = 4x + 3$$

$$f(x) = \frac{2x+7}{x+1}$$

- (i) State the value of x for which $f(x)$ is undefined. [1]

$$f(x) = \frac{2x+7}{x+1}$$

A function is undefined when the denominator is equal to zero.

When $x + 1 = 0$,

$$x = -1$$

$\therefore f(x)$ is undefined when $x = -1$.

- (ii) Calculate the value of $gf(5)$. [3]

Firstly,

$$\begin{aligned} f(5) &= \frac{2(5)+7}{5+1} \\ &= \frac{17}{6} \end{aligned}$$

Now,

$$\begin{aligned} gf(5) &= g\left(\frac{17}{6}\right) \\ &= 4\left(\frac{17}{6}\right) + 3 \\ &= \frac{34}{3} + 3 \\ &= \frac{43}{3} \end{aligned}$$

(iii) Find $f^{-1}(x)$. [3]

$$f(x) = \frac{2x+7}{x+1}$$

Let $y = f(x)$.

$$y = \frac{2x+7}{x+1}$$

Interchange variables x and y .

$$x = \frac{2y+7}{y+1}$$

Make y the subject.

$$x(y+1) = 2y+7$$

$$xy + x = 2y + 7$$

$$xy - 2y = 7 - x$$

$$y(x-2) = 7-x$$

$$y = \frac{7-x}{x-2}$$

$$\therefore f^{-1}(x) = \frac{7-x}{x-2}$$

(b) A ball is thrown vertically upwards. Its height, h metres, above the ground after t seconds is shown in the table below.

t (s)	0	1	2	3	4	5	6
h (m)	0	50	80	90	80	50	0

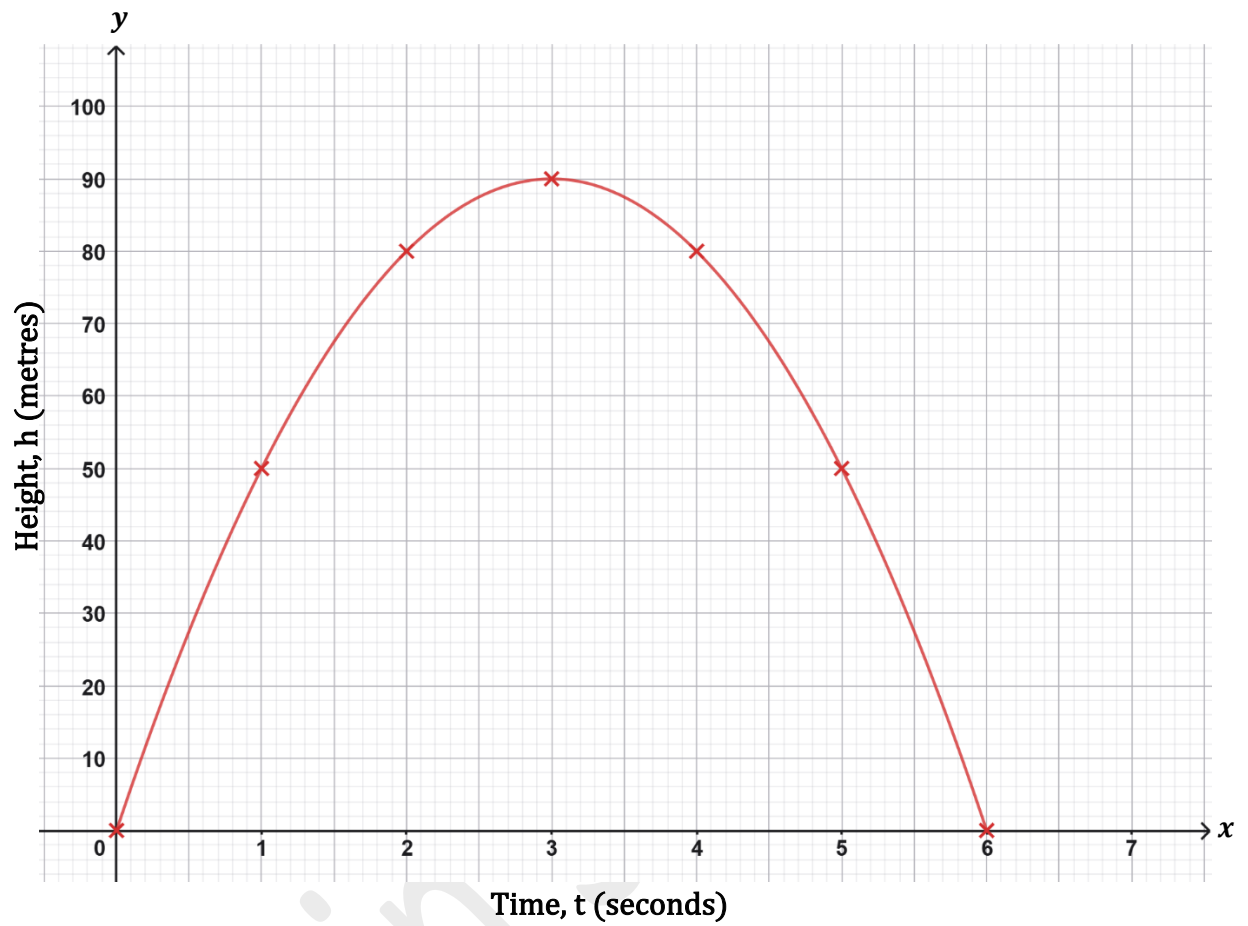
- (i) Using **2 cm to represent 1 second** on the x -axis and **1 cm to represent 10 metres** on the y -axis, plot a graph to show the height of the ball during the first 6 seconds. [4]

Scale Used:

x -axis: 2 cm represents 1 second

y -axis: 1 cm represents 10 metres

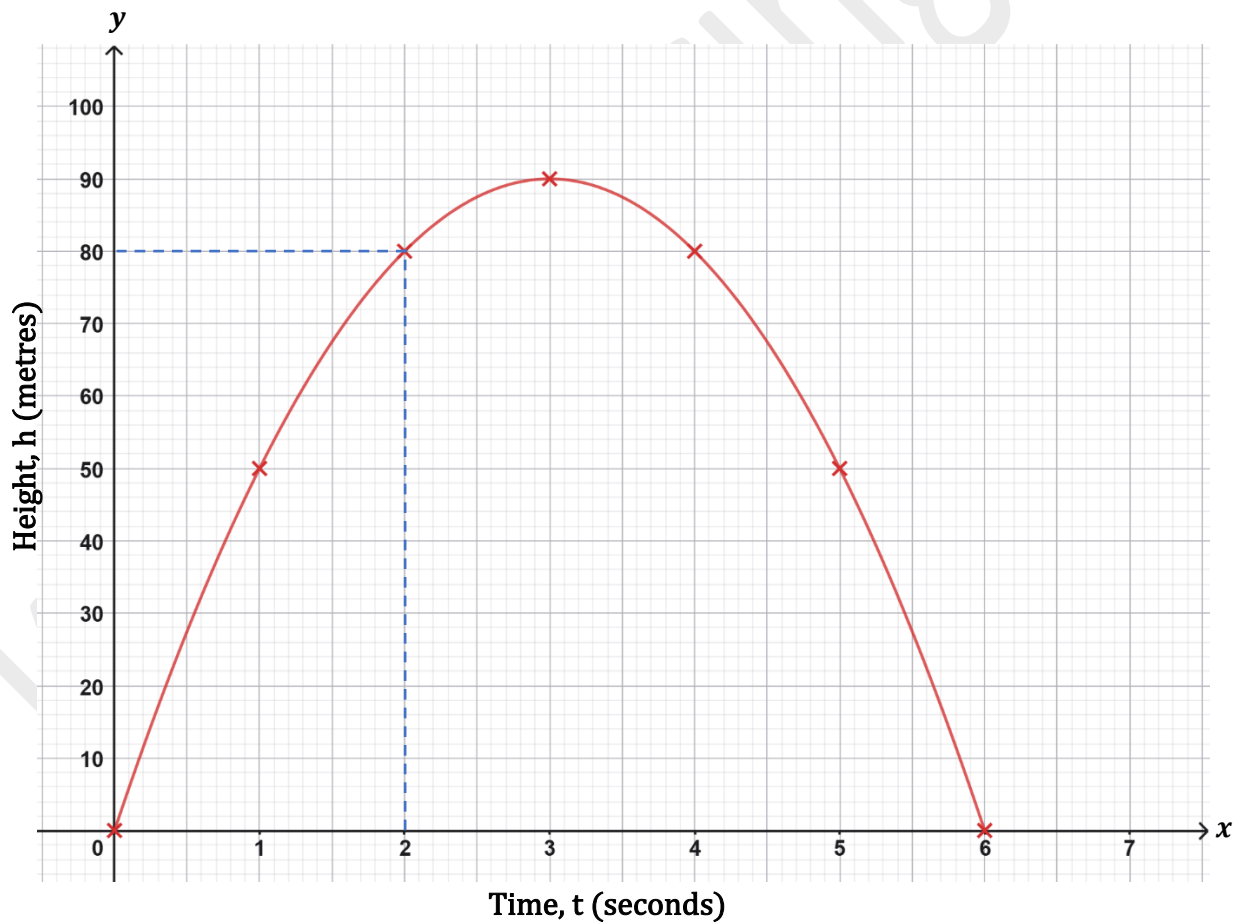
The graph of h VS t is shown on the next page:



(ii) Using your graph, determine

(a) the average speed of the ball during the first 2 seconds [2]

$$\begin{aligned}\text{Average speed} &= \frac{\text{Total distance covered}}{\text{Total time taken}} \\ &= \frac{80 \text{ m}}{2 \text{ s}} \\ &= 40 \text{ m s}^{-1}\end{aligned}$$



(b) the speed of the ball when $t = 3$ seconds.

The tangent to the graph when $t = 3$ is a horizontal line.

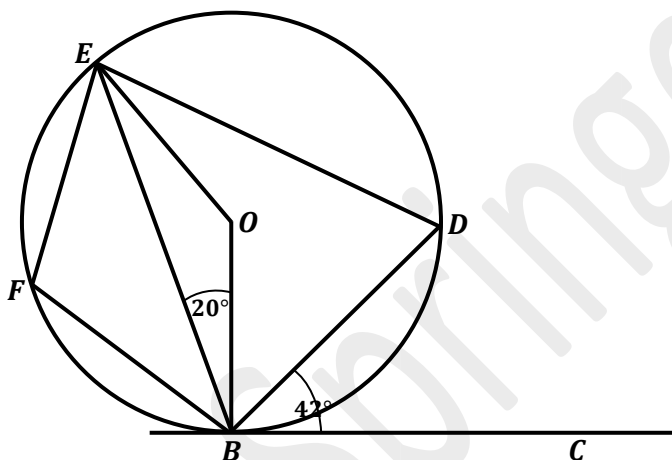
The gradient of a horizontal line is 0.

Therefore, the speed of the ball when $t = 3$ is 0 ms^{-1} .

Total: 15 marks

MEASUREMENT, GEOMETRY AND TRIGONOMETRY

10. (a) The diagram below, **not drawn to scale**, shows a circle, centre O . The line BC is a tangent to the circle at B . Angle $CBD = 42^\circ$ and angle $OBE = 20^\circ$.



Calculate, giving a reason for EACH step of your answer, the measure of:

- (i) $\angle BOE$ [2]

Since OB and OE are both radii of the same circle, then $OB = OE$.

Since the base angles of an isosceles triangle are equal, then $\angle OEB = 20^\circ$.

The sum of the interior angles of a triangle add up to 180° .

$$\therefore \angle BOE = 180^\circ - (20^\circ + 20^\circ)$$

$$= 180^\circ - 40^\circ$$

$$= 140^\circ$$

(ii) $\angle OED$

[2]

The angle made by a tangent to a circle and any chord, at the point of contact, is equal to the angle in the alternate segment of the circle.

$$\text{So, } \widehat{BED} = 42^\circ$$

$$\begin{aligned} \therefore \widehat{OED} &= 42^\circ - 20^\circ \\ &= 22^\circ \end{aligned}$$

(iii) $\angle BFE$

[3]

The angle subtended by a chord at the center of a circle is twice the angle that the chord subtends at the circumference, standing on the same arc.

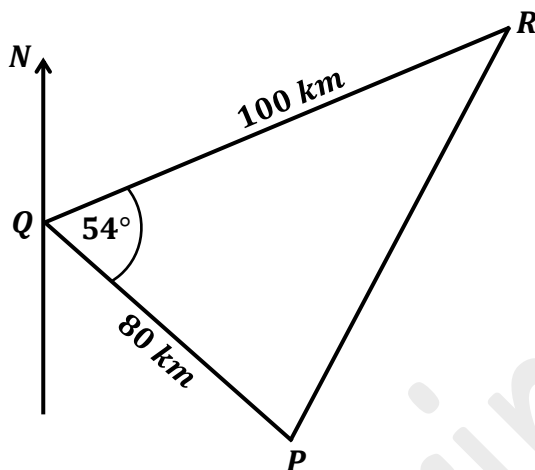
So,

$$\begin{aligned} \widehat{EDB} &= \frac{1}{2}(140^\circ) \\ &= 70^\circ \end{aligned}$$

The opposite angles of a cyclic quadrilateral are supplementary.

$$\begin{aligned} \therefore \widehat{BFE} &= 180^\circ - 70^\circ \\ &= 110^\circ \end{aligned}$$

- (b) The diagram below, **not drawn to scale**, shows the positions of three ports, P , Q and R .



Q is 80 km from P .

R is 100 km from Q on a bearing of 066° .

$\angle PQR = 54^\circ$.

Calculate

- (i) the bearing of P from Q

[2]

$$\begin{aligned}\text{The bearing of } P \text{ from } Q &= 66^\circ + 54^\circ \\ &= 120^\circ\end{aligned}$$

- (ii) the distance PR correct to 2 decimal places

[3]

Applying the cosine rule,

$$PR^2 = (100)^2 + (80)^2 - 2(100)(80) \cos 54^\circ$$

$$PR^2 = 6995.436$$

$$PR = \sqrt{6995.436}$$

$$PR = 83.64 \text{ km (to 2 decimal places)}$$

- (iii) the measure of $\angle QPR$ to the **nearest** degree

[3]

Applying the sine rule,

$$\frac{83.64}{\sin 54^\circ} = \frac{100}{\sin Q\hat{P}R}$$

$$\sin Q\hat{P}R = \frac{100 \sin 54^\circ}{83.64}$$

$$\sin Q\hat{P}R = 0.9673$$

$$Q\hat{P}R = \sin^{-1}(0.9673)$$

$$\therefore Q\hat{P}R = 75^\circ \quad (\text{to the nearest degree})$$

Total: 15 marks

VECTORS AND MATRICES

11. (a) The matrix M is defined as

$$M = \begin{pmatrix} 7 & 2 \\ p & -1 \end{pmatrix}$$

Determine the value of p for which the matrix M does NOT have an inverse. [2]

The matrix M is singular if $\det M = 0$.

$$\det M = (7 \times -1) - (2 \times p)$$

$$0 = -7 - 2p$$

$$2p = -7$$

$$p = -\frac{7}{2}$$

(b) Express the equations

$$4x - 2y = 0$$

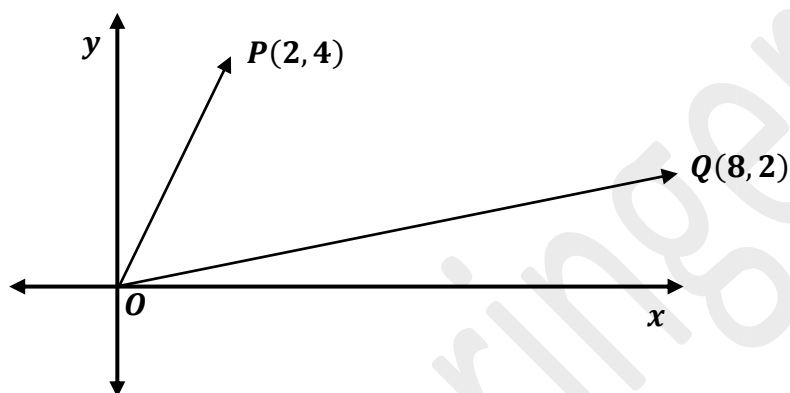
$$2x + 3y = 4$$

in the form $AX = B$, where A , X and B are matrices. [2]

Placing the given equation in the form $AX = B$ is

$$\begin{pmatrix} 4 & -2 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 4 \end{pmatrix}$$

- (c) In the diagram below, the coordinates of P and Q are $(2, 4)$ and $(8, 2)$ respectively. The line segment joining the origin $(0, 0)$ to the point P may be written as $\overrightarrow{OP}$.



- (i) What terms is used to describe $\overrightarrow{OP}$? [2]

OP is the position vector from O to P .

O is the fixed point from which the vector is drawn.

- (ii) Write EACH of the following in the form : $\begin{pmatrix} a \\ b \end{pmatrix}$

- (a) $\overrightarrow{OP}$ [1]

$P(2, 4)$

$\therefore OP = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$ which is of the form $\begin{pmatrix} a \\ b \end{pmatrix}$ where $a = 2$ and $b = 4$

(b) $\overrightarrow{OQ}$

[1]

$$Q(8, 2)$$

$$\therefore OQ = \begin{pmatrix} 8 \\ 2 \end{pmatrix} \quad \text{which is of the form } \begin{pmatrix} a \\ b \end{pmatrix} \text{ where } a = 8 \text{ and } b = 2$$

(c) $\overrightarrow{PQ}$

[2]

$$PQ = PO + OQ$$

$$PQ = -\begin{pmatrix} 2 \\ 4 \end{pmatrix} + \begin{pmatrix} 8 \\ 2 \end{pmatrix}$$

$$PQ = \begin{pmatrix} 6 \\ -2 \end{pmatrix} \quad \text{which is of the form } \begin{pmatrix} a \\ b \end{pmatrix} \text{ where } a = 6 \text{ and } b = -2$$

(iii) Given that $\overrightarrow{OP} = \overrightarrow{RQ}$, determine the coordinates of the point, R . [3]

Let R be the point (x, y) .

$$\text{So, } OR = \begin{pmatrix} x \\ y \end{pmatrix}.$$

$$RQ = RO + OQ$$

$$RQ = -\begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 8 \\ 2 \end{pmatrix}$$

$$RQ = \begin{pmatrix} -x + 8 \\ -y + 2 \end{pmatrix}$$

Since $OP = RQ$, then

$$\begin{pmatrix} 2 \\ 4 \end{pmatrix} = \begin{pmatrix} -x + 8 \\ -y + 2 \end{pmatrix}$$

Comparing the equivalent matrices, we get:

$$2 = -x + 8$$

$$x = 8 - 2$$

$$x = 6$$

$$4 = -y + 2$$

$$y = 2 - 4$$

$$y = -2$$

$\therefore OP = \begin{pmatrix} 6 \\ -2 \end{pmatrix}$ and the coordinates of R will be $(6, -2)$.

(iv) State the type of quadrilateral formed by $PQRO$.

Justify your answer.

[2]

The quadrilateral $PQRO$ is a parallelogram.

Reason:

$$\overrightarrow{OP} = \overrightarrow{RQ}$$

So one pair of opposite sides are equal and parallel.

Therefore, $PQRO$ is a parallelogram.

Total: 15 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.